

# bayes\_model

2026-03-23

```
colorectal_data <- read.csv('colorectal_cancer_dataset.csv')
head(colorectal_data)
```

|      | Patient_ID              | Country                    | Age              | Gender            | Cancer_Stage      | Tumor_Size_mm | Family_History |
|------|-------------------------|----------------------------|------------------|-------------------|-------------------|---------------|----------------|
| ## 1 | 1                       | UK                         | 77               | M                 | Localized         | 69            | No             |
| ## 2 | 2                       | UK                         | 59               | M                 | Localized         | 33            | No             |
| ## 3 | 3                       | Japan                      | 66               | M                 | Regional          | 17            | No             |
| ## 4 | 4                       | USA                        | 83               | M                 | Regional          | 14            | No             |
| ## 5 | 5                       | France                     | 66               | M                 | Localized         | 34            | No             |
| ## 6 | 6                       | Japan                      | 69               | M                 | Localized         | 71            | No             |
|      | Smoking_History         | Alcohol_Consumption        | Obesity_BMI      | Diet_Risk         | Physical_Activity |               |                |
| ## 1 | No                      |                            | Yes Overweight   | Low               |                   |               | Low            |
| ## 2 | No                      |                            | No Overweight    | Moderate          |                   |               | Low            |
| ## 3 | Yes                     |                            | No Normal        | Low               |                   |               | Low            |
| ## 4 | No                      |                            | No Obese         | High              |                   |               | Low            |
| ## 5 | Yes                     |                            | No Normal        | Low               |                   |               | Moderate       |
| ## 6 | No                      |                            | Yes Obese        | Moderate          |                   |               | Low            |
|      | Diabetes                | Inflammatory_Bowel_Disease | Genetic_Mutation | Screening_History |                   |               |                |
| ## 1 | No                      |                            | No               | No                |                   |               | Regular        |
| ## 2 | No                      |                            | No               | No                |                   |               | Regular        |
| ## 3 | No                      |                            | Yes              | No                |                   |               | Irregular      |
| ## 4 | No                      |                            | No               | No                |                   |               | Regular        |
| ## 5 | No                      |                            | Yes              | No                |                   |               | Never          |
| ## 6 | Yes                     |                            | Yes              | No                |                   |               | Regular        |
|      | Early_Detection         | Treatment_Type             | Survival_5_years | Mortality         | Healthcare_Costs  |               |                |
| ## 1 | Yes                     | Combination                | Yes              | No                |                   |               | 54413          |
| ## 2 | No                      | Chemotherapy               | Yes              | No                |                   |               | 76553          |
| ## 3 | No                      | Chemotherapy               | Yes              | No                |                   |               | 62805          |
| ## 4 | No                      | Surgery                    | Yes              | No                |                   |               | 89393          |
| ## 5 | Yes                     | Surgery                    | Yes              | No                |                   |               | 66425          |
| ## 6 | No                      | Radiotherapy               | No               | No                |                   |               | 82764          |
|      | Incidence_Rate_per_100K | Mortality_Rate_per_100K    | Urban_or_Rural   |                   |                   |               |                |
| ## 1 | 50                      | 5                          | Urban            |                   |                   |               |                |
| ## 2 | 37                      | 25                         | Urban            |                   |                   |               |                |
| ## 3 | 54                      | 27                         | Urban            |                   |                   |               |                |
| ## 4 | 45                      | 11                         | Urban            |                   |                   |               |                |
| ## 5 | 15                      | 27                         | Urban            |                   |                   |               |                |
| ## 6 | 41                      | 19                         | Urban            |                   |                   |               |                |
|      | Economic_Classification | Healthcare_Access          | Insurance_Status |                   |                   |               |                |
| ## 1 | Developed               | Moderate                   | Insured          |                   |                   |               |                |
| ## 2 | Developing              | High                       | Uninsured        |                   |                   |               |                |
| ## 3 | Developed               | Moderate                   | Uninsured        |                   |                   |               |                |
| ## 4 | Developed               | Moderate                   | Insured          |                   |                   |               |                |
| ## 5 | Developing              | High                       | Insured          |                   |                   |               |                |

```
## 6          Developed          Moderate          Uninsured
## Survival_Prediction
## 1          Yes
## 2          Yes
## 3          No
## 4          Yes
## 5          Yes
## 6          No
```

```
# Determine tumor size based on patient data
subset_overall <- colorectal_data[, c(
  "Healthcare_Costs",
  "Economic_Classification",
  "Healthcare_Access",
  "Insurance_Status"
)]

library(MCMCpack)
```

```
## Loading required package: coda
```

```
## Loading required package: MASS
```

```
## ##
## ## Markov Chain Monte Carlo Package (MCMCpack)
```

```
## ## Copyright (C) 2003-2026 Andrew D. Martin, Kevin M. Quinn, and Jong Hee Park
```

```
## ##
## ## Support provided by the U.S. National Science Foundation
```

```
## ## (Grants SES-0350646 and SES-0350613)
## ##
```

```
library(mvnfast)
y <- subset_overall$Healthcare_Costs

X <- model.matrix(Healthcare_Costs ~ factor(Economic_Classification) +
                  factor(Healthcare_Access) +
                  factor(Insurance_Status),
                  data = subset_overall)

p <- ncol(X)
prior_w <- matrix(0, nrow = p, ncol = 1)
prior_Sigma <- diag(p) * 1e6
y <- as.matrix(y)
X <- as.matrix(X)
n <- length(y)
library(MCMCpack)

# Now need priors for sigma_y
prior_alpha <- 3
```

```

prior_beta <- var(y) * (prior_alpha - 1)
# will need the sample size
n <- length(y)

# Set up markov chain matrices
no_samples <- 1000
gibbs_samples_w <- matrix(0, nrow = no_samples, ncol = p)
gibbs_samples_sigmay <- matrix(0, nrow = no_samples, ncol = 1)

#initialize
ols_fit <- lm(Healthcare_Costs ~ factor(Economic_Classification) +
              factor(Healthcare_Access) +
              factor(Insurance_Status),
              data = subset_overall)
w_tilde <- matrix(coef(ols_fit), ncol = 1)
sigma2_tilde <- summary(ols_fit)$sigma^2

# run the sampler
for(i in 1:no_samples){
  # conditional posterior parameters for w
  V = solve((1/sigma2_tilde)*t(X)%*%X + solve(prior_Sigma))

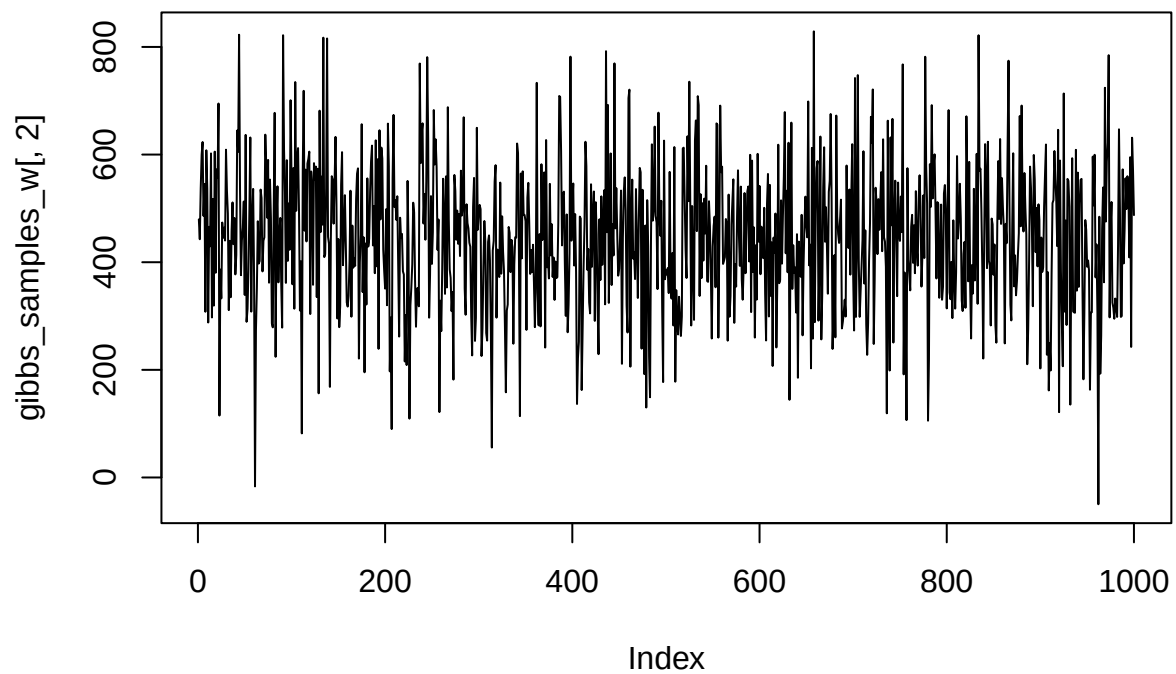
  M = V%*%((1/sigma2_tilde)*t(X)%*%y + solve(prior_Sigma)%*%prior_w)
  # sample weights
  w_tilde <- t(rmvn(1, M, V)) # rmvn gives w^T

  # conditional posterior parameters for sigma2
  A = (2*prior_alpha + n)/2
  B = (2*prior_beta + t(y - X%*%w_tilde)%*%(y - X%*%w_tilde))/2
  # sample sigma2
  sigma2_tilde <- max(rinvgamma(1, A, B), 1e-6)

  # store them
  gibbs_samples_w[i,] <- w_tilde
  gibbs_samples_sigmay[i,] <- sigma2_tilde
}

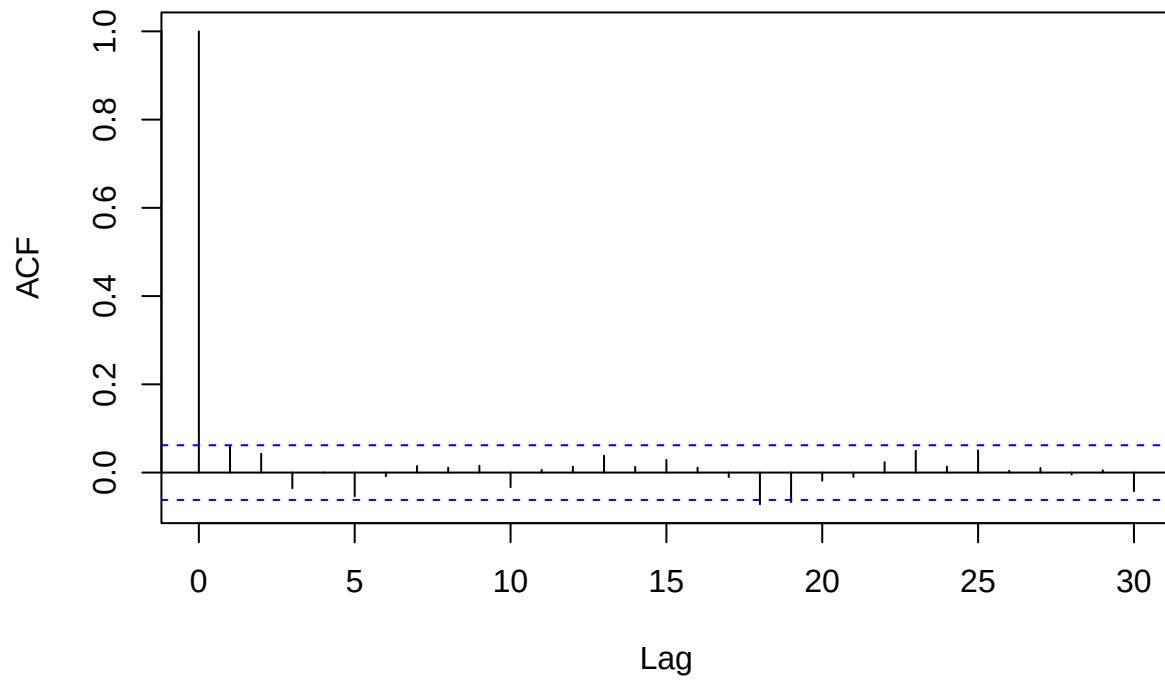
# check convergence and mixing of weights
plot(gibbs_samples_w[,2], type = "l") # change the index to examine the 7 weights

```

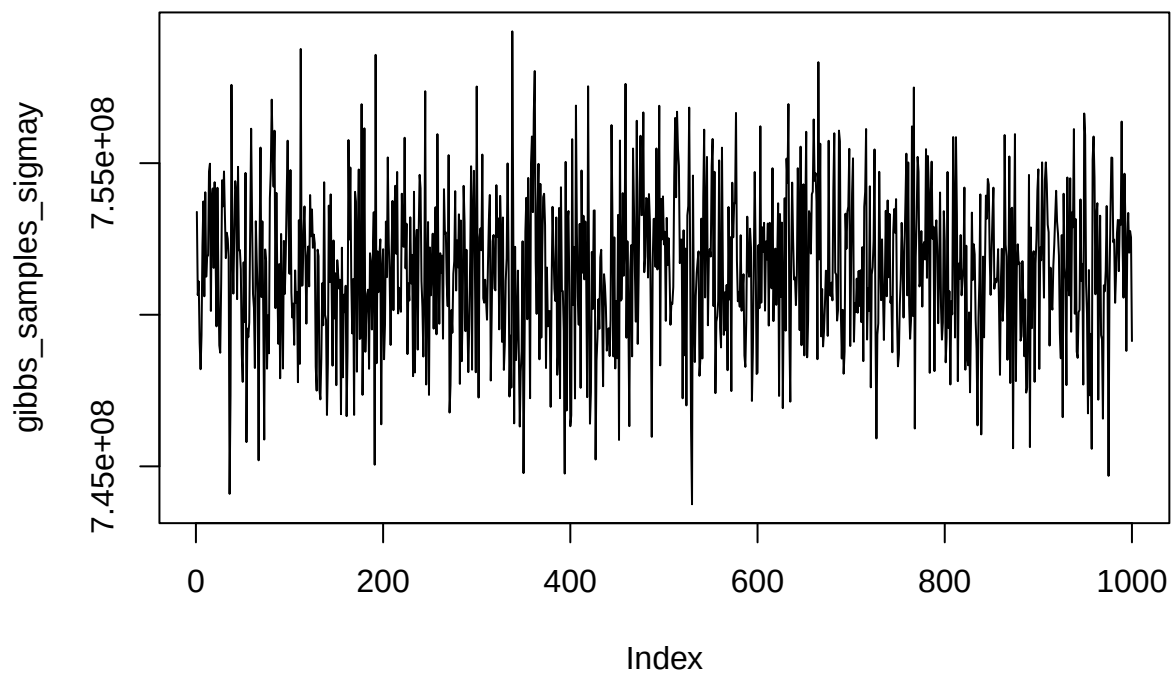


```
acf(gibbs_samples_w[,2]) # change the index to examine the 7 weights
```

### Series gibbs\_samples\_w[, 2]

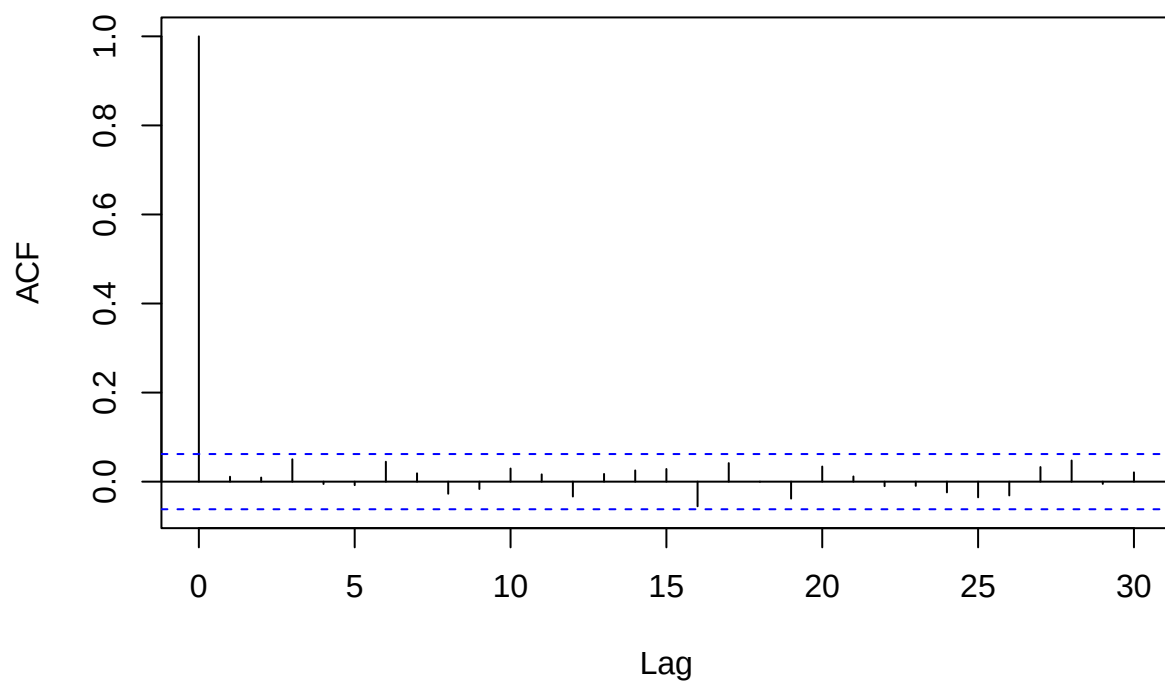


```
# check convergence and mixing of variance term  
plot(gibbs_samples_sigmay, type = "l")
```



```
acf(gibbs_samples_sigmay)
```

## Series 1

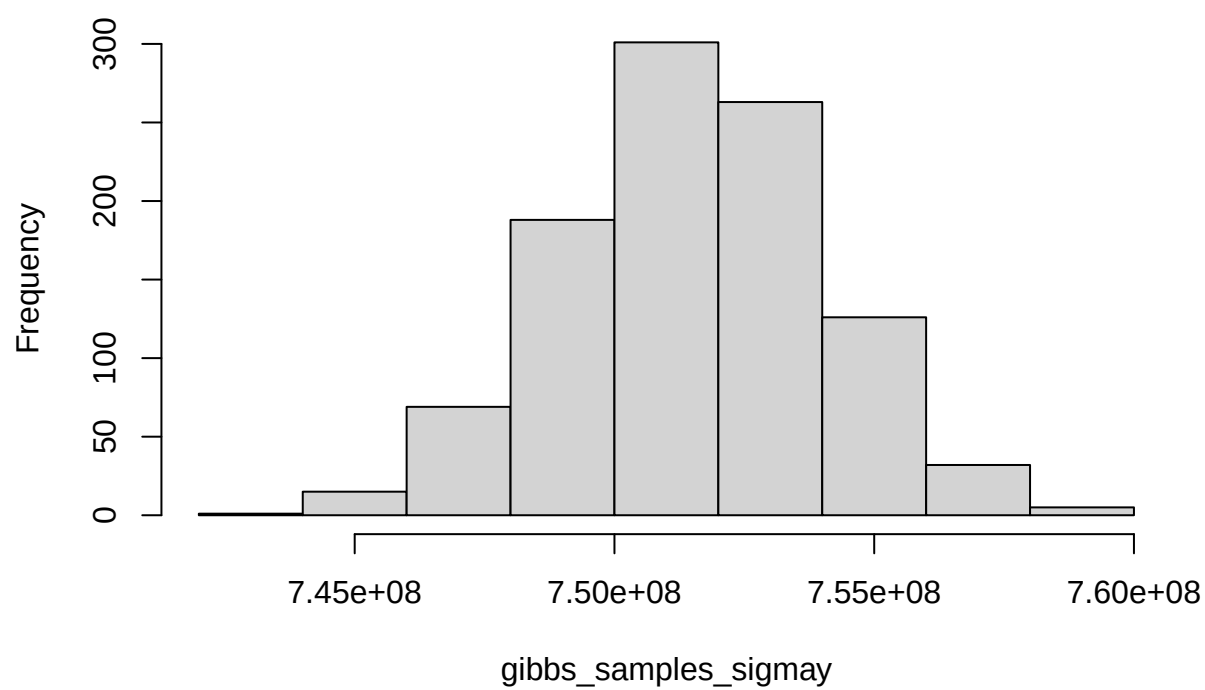


```
# converged quickly, very good mixing. Not even going to bother with burn-in or thinning  
# though, you may need to in practice
```

```
# visualize posteriors?
```

```
hist(gibbs_samples_sigmay, main = "Posterior of Variance of Health Care Costs")
```

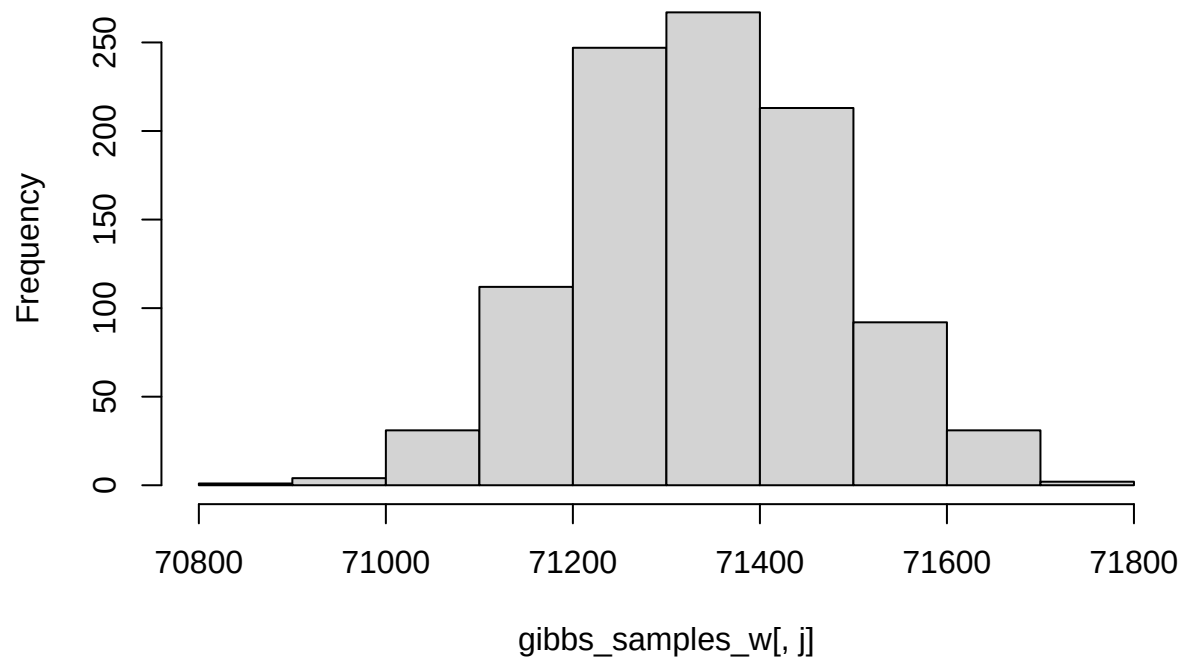
## Posterior of Variance of Health Care Costs

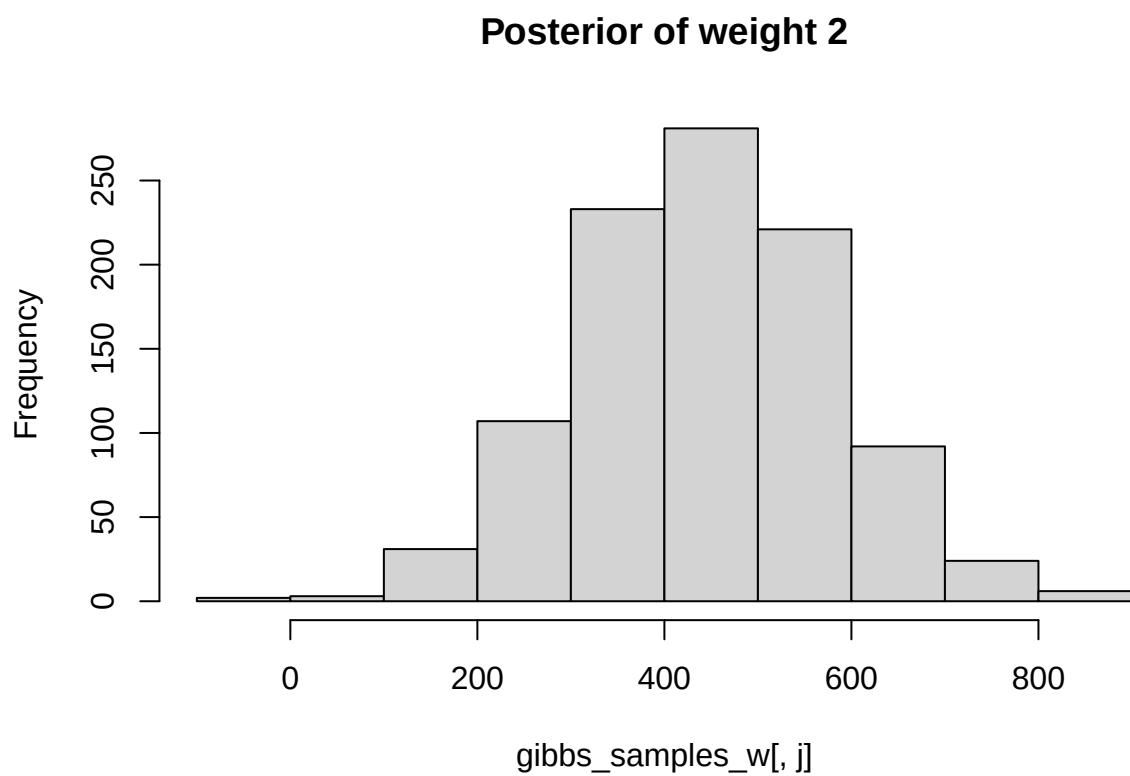


```
for (j in 1:p) {  
  hist(gibbs_samples_w[,j], main = paste("Posterior of weight", j))  
}
```

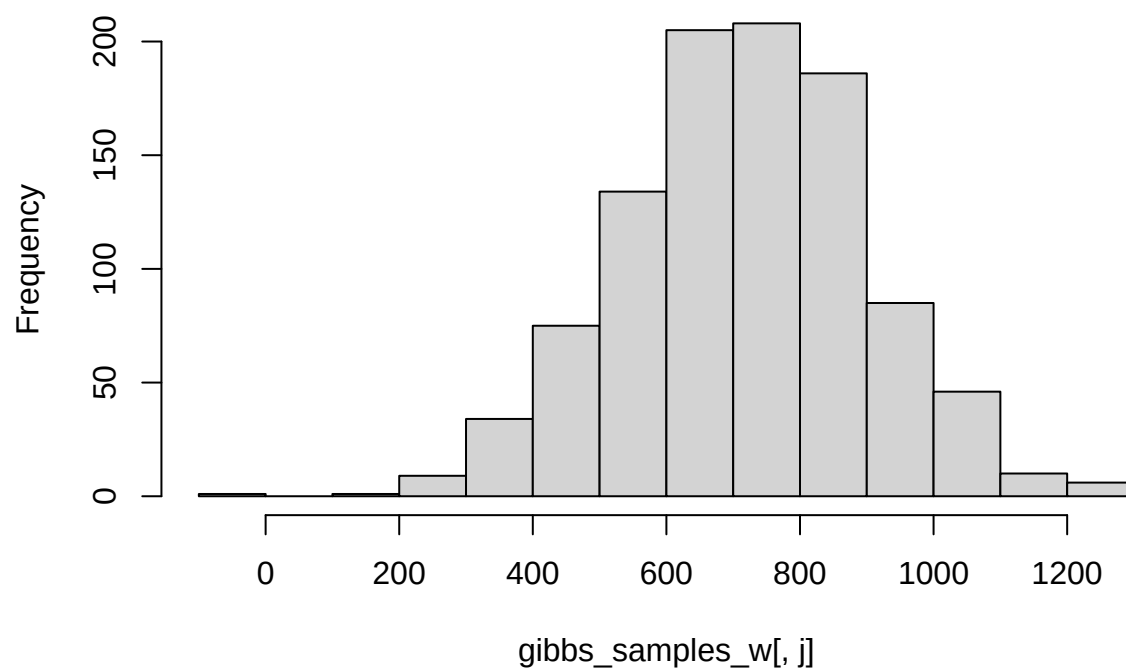


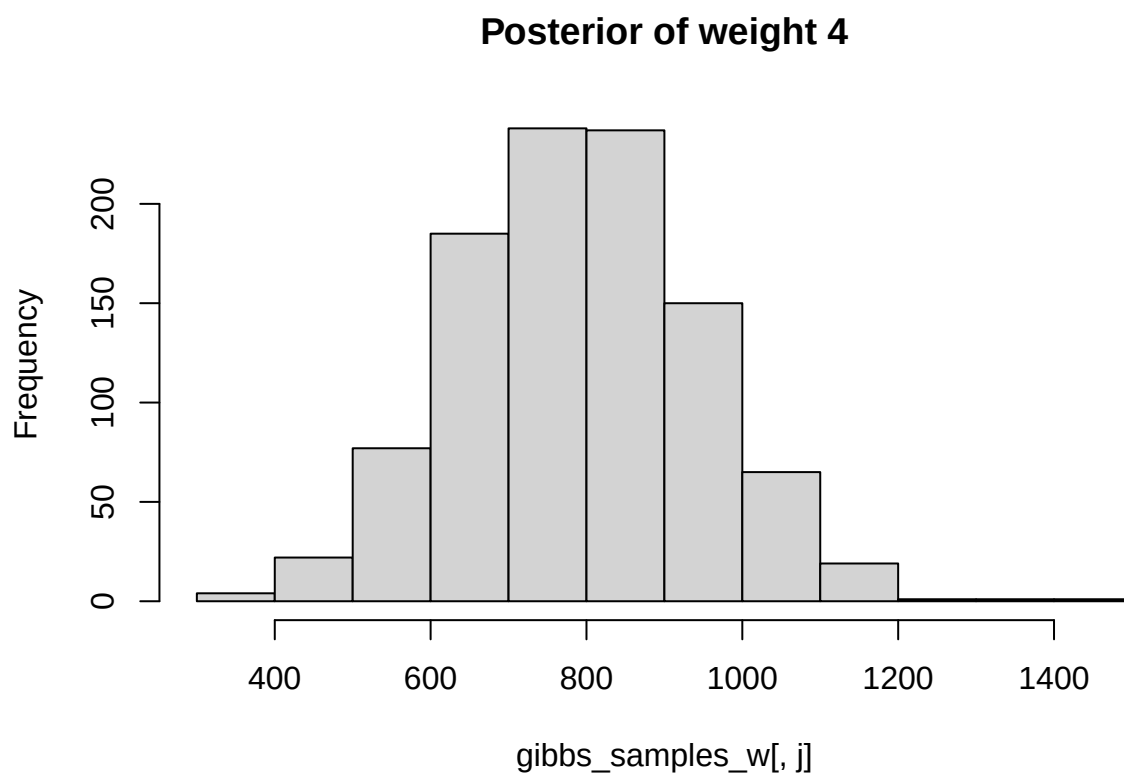
**Posterior of weight 1**



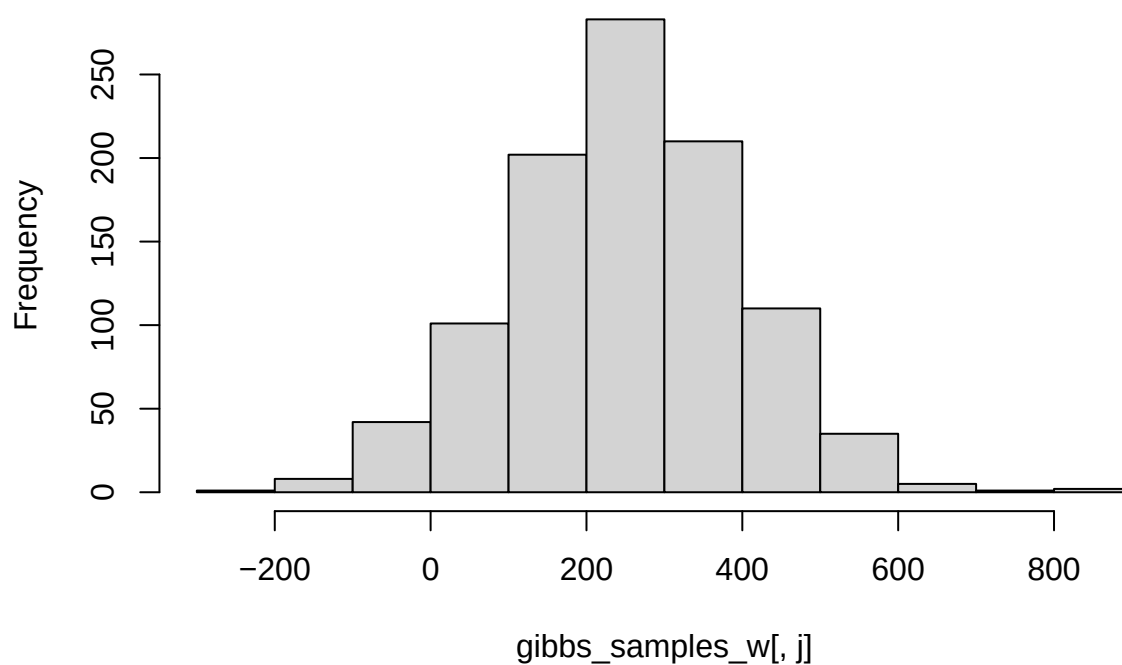


**Posterior of weight 3**



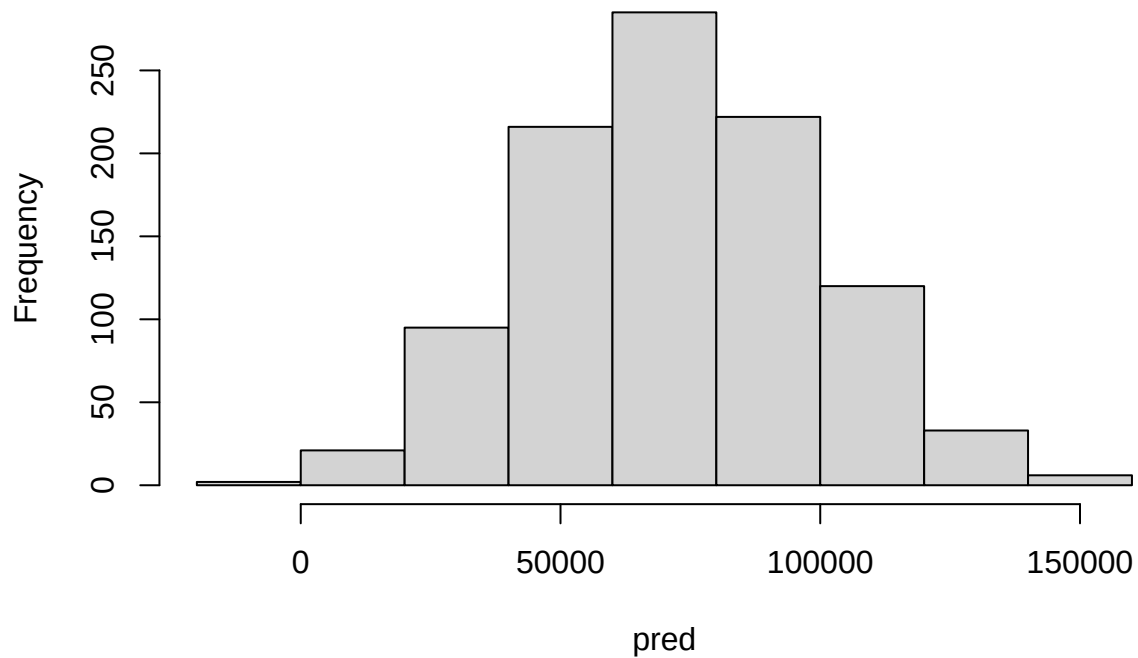


## Posterior of weight 5



```
# Generating predictive distribution for Healthcare
pred <- numeric(nrow(gibbs_samples_w))
for(i in 1:nrow(gibbs_samples_w)){
  pred[i] <- rnorm(1, mean = t(gibbs_samples_w[i,]) %*% X[1,], sd = sqrt(gibbs_samples_sigmay[i]))
}
hist(pred, main = "Posterior Predictive Distribution")
```

## Posterior Predictive Distribution



```
cat("y mean:", mean(y), "\n")           # should be ~66,000
```

```
## y mean: 72452.17
```

```
cat("pred mean:", mean(pred), "\n")
```

```
## pred mean: 72440.92
```

```
cat("y range:", range(y), "\n")         # should be 25,033 to 119,469
```

```
## y range: 25000 119999
```

```
cat("X dimensions:", dim(X), "\n")      # should be 100 x p
```

```
## X dimensions: 167497 5
```

```
cat("p:", p, "\n")
```

```
## p: 5
```

```
cat("prior_beta:", prior_beta, "\n")
```

```
## prior_beta: 1501824659
```

```
cat("95% Credible Intervals for each weight:\n")
```

```
## 95% Credible Intervals for each weight:
```

```
cat(sprintf("%-15s %-15s %-15s\n", "Weight", "Lower (2.5%)", "Upper (97.5%)"))
```

```
## Weight          Lower (2.5%)    Upper (97.5%)
```

```
cat(rep("-", 45), "\n")
```

```
## - - - - -
```

```
for (j in 1:p) {  
  lower <- quantile(gibbs_samples_w[, j], 0.025)  
  upper <- quantile(gibbs_samples_w[, j], 0.975)  
  cat(sprintf("%-15s %-15.4f %-15.4f\n",  
              colnames(X)[j], lower, upper))  
}
```

```
## (Intercept)      71082.9759      71620.1671  
## factor(Economic_Classification)Developing 177.9157      709.0040  
## factor(Healthcare_Access)Low 353.5867      1067.8510  
## factor(Healthcare_Access)Moderate 492.9206      1087.4655  
## factor(Insurance_Status)Uninsured -41.3177      537.5275
```

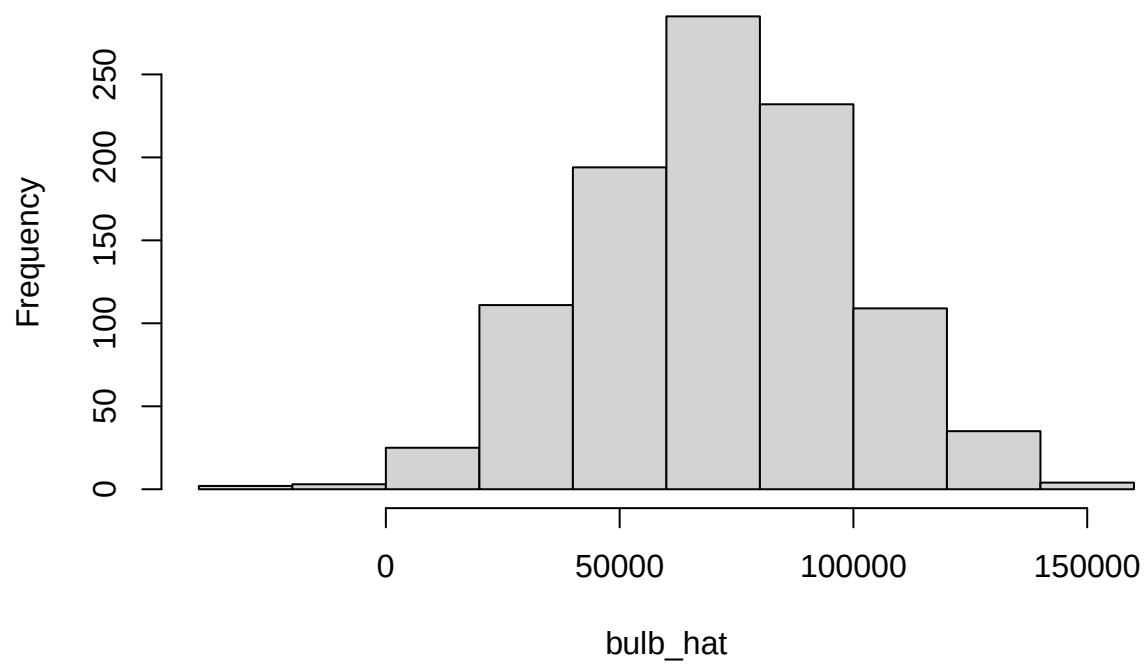
```
# Generating predictive distribution for Bulbasaur
```

```
bulb_hat <- c()
```

```
for(i in 1:nrow(gibbs_samples_w)){  
  bulb_hat <- c(bulb_hat, rnorm(n = 1, mean = t(gibbs_samples_w[i,])%*%X[1,], sd = sqrt(gibbs_samples_s
```

```
hist(bulb_hat, main = "Post Pred Dist of the first row")
```

### Post Pred Dist of the first row



```
mean(bulb_hat)
```

```
## [1] 71199.01
```

```
y[1]
```

```
## [1] 54413
```